INCH-POUND
MIL-B-44381 (GL)
5 February 1990

MILITARY SPECIFICATION

BOARD, MAP, COMPLEXING (SICPS)

This specification is approved for use by the Natick Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a complexable map board.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-375	-	Plastic Film, Flexible, Vinyl Chloride
NN-P-530	-	Plywood, Flat Panel
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7195

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-4-7457 - Mapboard Assembly
5-4-7461 - Mapboard

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the documents cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3935 - Polycarbonate (PC) Unfilled & Reinforced Materials
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. Materials and components shall be as specified on Drawing 5-4-7457 and on all drawings and parts lists pertaining thereto, and as specified herein. Materials not definitely specified shall be of the quality normally used by the contractor provided that the completed item complies with all the provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Overlays. The plastic overlay shall conform to type II, class 1 (12 gauge double polished, clear, 2S) of L-P-375. The polycarbonate overlay shall conform to ASTM D 3935 (PC 11B359 (800 J/M) 45Z01, 1/16-inch sheet, clear) (see 4.4.1.1).

3.3 Design and construction. The design and construction of the map board shall be as specified herein and conform to Drawing 5-4-7457 and all subsidiary drawings and parts lists. The map boards shall be capable of being complexed together by hand without the use of a tool for force.

3.3.1 Screws. All screws shall be securely set.

3.4 Finish. Plywood shall be painted on both sides and edges as specified on Drawing 5-4-7461.

3.5 Map board identification. The back of each map board section shall be permanently and distinctly stamped or marked for identification in accordance with MIL-STD-130, with the letters "US" in 1/4-inch high characters, and the manufacturer's name, number and date of the contract in 1/8-inch characters.

3.6 Workmanship. The map board shall conform to the quality of product established by this specification. All components of the map board shall not be bent out of shape and shall be free from split, puncture, splinter, scratch, rip, fracture, twist, malformation, sharp edge, and burr.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the plastic and polycarbonate overlays are as specified in 3.2.1.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of map boards. The sample unit shall be one map board. The presence of any major defect shall be cause for rejection of the map board. The inspection level shall be II. An acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5 for minor defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Color not as specified		201
Hardware	Broken, malformed	101	
	Locators do not rotate freely		202
	Corroded area		203
	Burr or sharp edge which may cause injury	102	
Construction and Workmanship	Overlays do not fit onto map board locators	103	
	Overlays extend past the edges of the map board	104	
	Overlays do not extend to the edge of the map board squarely	105	
	Any part or component not fabricated of applicable referenced materials	106	
	Any component missing	107	
	Any component split, punctured, twisted, fractured, malformed, splintered, ripped, scratched, or bent out of shape	108	
	Any sharp edge or burr	109	
Threaded components	Any bolt, nut or screw missing	110	
	Not specified size or type	111	
	Threads damaged or stripped	112	
Markings and Labels	Omitted, illegible, incorrect, or improper size or type, or not in location specified		204

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerances shall be classified as a defect. The lot size shall be expressed in units of map boards. The sample unit shall be one map board. The inspection level shall be S-2. The presence of any defect shall be cause for rejection of the lot.

4.4.5 End item testing. The end items shall be tested for the characteristics listed in table II. The lot size shall be expressed in terms of map boards. The sample unit shall be one map board. The inspection level shall be II. Any test failure shall be cause for rejection of the lot.

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TABLE II. End item tests

Examine	Defect
Assembly	Component is out of alignment Locators do not swivel Map boards do not easily and quickly lock together to any other random map board Plastic and polycarbonate overlays which do not easily fit over locators

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2. The AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each map board shall be placed within a close fitting fiberboard box conforming to style OPF, type CF, grade 200 of PPP-B-636. The inside dimensions of each box shall approximate 60-1/2 inches by 27 inches by 2-3/4 inches. Each box shall be closed in accordance with PPP-B-636.

5.1.2 Commercial preservation. Components of each frame shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

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5.2.1 Level A packing. Ten map boards, preserved as specified in 5.1, shall be packed in a close fitting, cleated plywood shipping container conforming to overseas type, style A of PPP-B-601. Inside dimensions of the box shall approximate 61-1/2 inches in length, 28 inches in width, and 27-1/2 inches in depth. Each shipping container shall be skidded, closed and reinforced with strapping in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Ten map boards, preserved as specified in 5.1, shall be packed in a close fitting, cleated plywood shipping container conforming to domestic type, style optional of PPP-B-601. Inside dimensions of the box shall approximate 61-1/2 inches in length, 28 inches in width, and 27-1/2 inches in depth. Each shipping container shall be skidded, closed and reinforced with strapping in accordance with the appendix of PPP-B-601.

5.2.3 Commercial packing. Map boards preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The map board shall provide a smooth flat surface to layout maps, drawings, bulletins, etc.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 Subject term (key word) listing.

Bulletin
Drawing
Map
SICPS
Tent

Custodian:

Army - GL

Preparing activity:

Army - GL

(Project 7195-A105)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-B-44381 (GL)	Board, Map , Complexing

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements

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